

user experience (UX). The latest AI advancements have allowed designers to create a more immersive and personalized online experience for customers. These AI tools have revolutionized modern web design, establishing standard components of human-machine interaction. One such example is the widespread use of chatbots, which can be found on almost every website and mobile app. Through complex machine learning procedures and advanced AI algorithms, chatbots are bridging the gap between human cognition and machine automation. As a result, interfaces are becoming more engaging, responsive, and ultimately more human-like.

Chatbots powered by AI are projected to save online businesses \$8 billion in business costs by 2022. The influence of AI-based applications has transformed traditional web design and significantly altered user interaction with websites. AI-driven chatbots have revolutionized communication by generating human-like responses that eliminate the need for pre-programmed conversational pathways. Advanced natural language processing (NLP) systems enable chatbots to initiate meaningful conversations and drive engagement with users.

The Watson Assistant is an advanced AI chatbot that is capable of creating its own knowledge base by analyzing a business's historical chat logs and calls. Additionally, it can prompt customers for more information, request human assistance, suggest conversation styles, and autonomously manage messaging channels, mobile apps, and service tools on a website.

The real estate industry has recorded the highest usage of AI chatbots, with 28%, followed by travel at 16% and education at 14%. According to a report by Gartner, AI chatbots will handle 85% of customer interactions by 2020. By 2023, chatbots will save a total of 2.3 billion hours for businesses and consumers. The overall market size of chatbots is expected to reach \$1.3 billion by 2024, according to a report by Global Market Insights.

AI for UI Improvements

Likewise, Deep learning and adaptive AI optimize User Interfaces (UIs), allowing them to make design decisions autonomously. This involves training the program to comprehend design principles and analyze existing pages of

